Jakub Kołodziej

Software Engineer in Mission Control 🛰️

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Working Experience

Mission Control Software Engineer - Creotech Instruments S.A., Warsaw *08/2025 - present*

- Developed the internal Mission Control System in Rust from the ground up, using the modern tokio async stack, according to CCSDS.
- Contributed to projects PIAST and Poland's highest-value satellite defense contract MIKROGLOB.
- Executed in-orbit Mission Control operations, as well as on-ground satellite integration tests.
- Maintained internal tools and critical Mission Control software in Python, pytest, django REST framework, Ansible, and Docker.
- Integrated external software from third-party suppliers.

IT System Administrator - Cema Mystal sp. z o.o., Myszków *11/2023 - 07/2025*

- Managed and monitored over 40 users and endpoints via Zabbix and Wazuh.
- Extended the company ERP system beyond vendor capabilities, implementing custom workflows for internal operations.
- Developed internal tools for data-extraction and processing pipelines: Python, Qt, SQL.
- Deployed and administered: Proxmox VMs, Windows Server, and SQL Server instances.

Skills

Category	Skills
Languages & Frameworks	Python 🐍(asyncio, django, django REST framework, pytest, scikit-learn), Rust 🦀 (tokio, axum, serde)
Databases & queues	NATS, MQTT, PostgreSQL (TimescaleDB), SQL Server, SQLite3
Infrastructure	Docker, Linux (RHEL deriatives), Ansible, Nix, Proxmox, Grafana (LGTM Stack),
Tools	git, uv, poetry, Gitlab CI, commitizen, Gitlab CI, Zed (Builtin Codex support)
Domain	REST API, OpenAPI, Event Driven Architecture, CCSDS, PUS, Mission Control, ECSS

Education & Qualifications

Bachelor of Engineering in IT - University of Silesia, *2021 - 2025*

- **Thesis**: *Device monitoring and log management system*: repository
- **Thesis grade**: 5, GPA: 3.5 (≈ 4.5 PL)

Projects

Language	Personal Projects	OSS Contributions - Projects: 8 - PRs: 9
Rust 🦀	minihashcat, ccsds_parsers	spacepackets-rs, RustScan, ncspot
Other 🧩	dotfiles (🏠, ❄️), sklearn_minisom (🐍)	Cephalon Navis (🎯), Skyblocker (☕), commitizen-docker (🐋)

Misc. Information

- **Languages**: English (B2), Polish (fluent/native)
- **Availability**: 1 month

GDPR / RODO (PL)

I agree to the processing of personal data provided in this document for realising the recruitment process pursuant to the Personal Data Protection Act of 10 May 2018 (Journal of Laws 2018, item 1000) and in agreement with Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation).